

TM Forum Component

Fault Management

TMFC043

Maturity Level: Alpha	Team Approved Date: 18-Jun-2026
Release Status: Preview	Approval Status: Team Approved
Version 1.1.0	IPR Mode: RAND

Notice

Copyright © TM Forum 2026. All Rights Reserved.

This document and translations of it may be copied and furnished to others, and derivative works that comment on or otherwise explain it or assist in its implementation may be prepared, copied, published, and distributed, in whole or in part, without restriction of any kind, provided that the above copyright notice and this section are included on all such copies and derivative works. However, this document itself may not be modified in any way, including by removing the copyright notice or references to TM FORUM, except as needed for the purpose of developing any document or deliverable produced by a TM FORUM Collaboration Project Team (in which case the rules applicable to copyrights, as set forth in the [TM FORUM IPR Policy](#), must be followed) or as required to translate it into languages other than English.

The limited permissions granted above are perpetual and will not be revoked by TM FORUM or its successors or assigns.

This document and the information contained herein is provided on an “AS IS” basis and TM FORUM DISCLAIMS ALL WARRANTIES, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY WARRANTY THAT THE USE OF THE INFORMATION HEREIN WILL NOT INFRINGE ANY OWNERSHIP RIGHTS OR ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

Direct inquiries to the TM Forum office:

100 Enterprise Drive
Suite 301 #1649
Rockaway, NJ 070866, USA
Tel No. +1 862 227 1648
TM Forum Web Page: www.tmforum.org

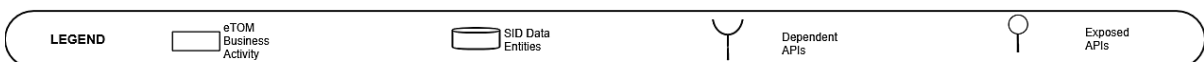
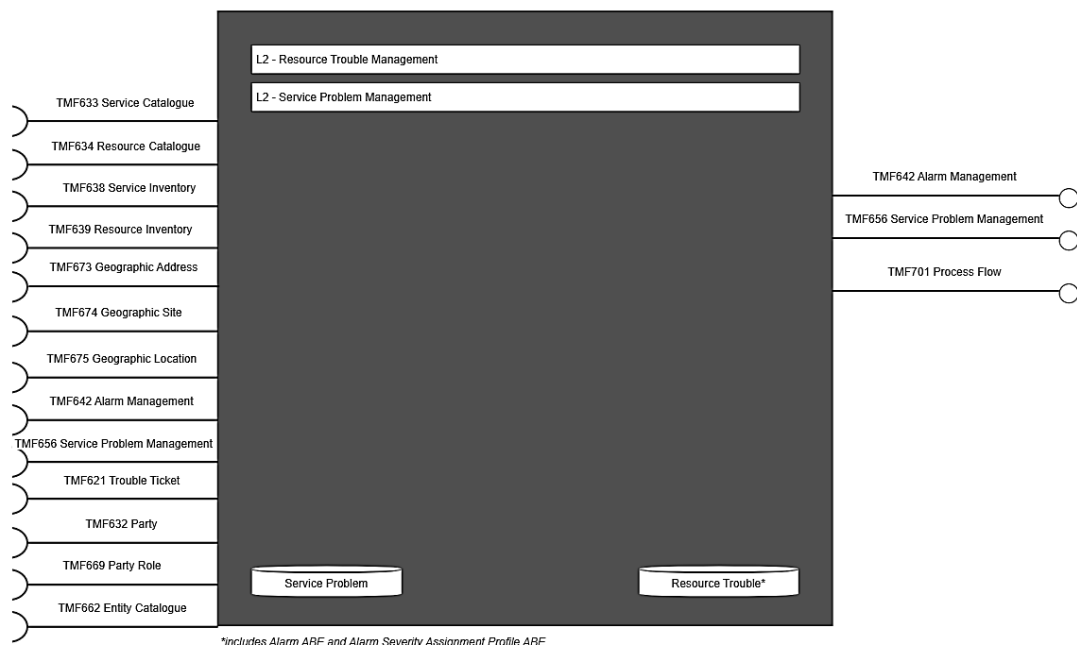
Table of Contents

Notice	2
Table of Contents	3
1. Overview	4
2. eTOM Processes, SID Data Entities and Functional Framework Functions	
5	
2.1. eTOM business activities	5
2.2. SID ABEs	5
2.3. eTOM L2 - SID ABEs links	5
2.4. Functional Framework Functions	6
3. TM Forum Open APIs & Events	7
3.1. Exposed APIs	8
3.2. Dependent APIs	11
3.3. Events	13
4. Machine Readable Component Specification.....	14
5. References	14
5.1. TMF Standards related versions	14
5.2. Jira References	14
5.2.1. SID	14
5.3. Further resources	14
6. Administrative Appendix.....	15
6.1. Document History.....	15
6.1.1. Version History.....	15
6.1.2. Release History.....	15
6.2. Acknowledgements.....	15

1. Overview

Component Name	ID	Description	ODA Function Block
Fault Management	TMFC043	Fault Management component involves identifying, diagnosing, and resolving issues that prevent the communication resources and their overlaying services from functioning correctly. Its scope covers this functionality at the ODA Production layer while business related resolutions are not part of its scope. The Fault Management component manages the currently active alarms or problem candidates, potentially modifying or enhancing their data to make persistent alarms more actionable. It may apply logic to preserve alarms that reflect actual problems, eliminating those that are merely symptoms. Various actions can be taken using the alarm/problem information as part of the resolution process, such as applying automated corrective actions or opening trouble tickets.	Production

TMF043: Fault Management



2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM business activities

eTOM business activities this ODA Component is responsible for.

Identifier	Level	Business Activity Name	Description
1.5.8	L2	Resource Trouble Management	Resource Trouble Management processes are responsible for the management of troubles, including security events, associated with specific resources. The objectives of these processes are to efficiently and effectively manage reported resource trouble, isolate the root cause and act to resolve the resource trouble.
1.4.6	L2	Service Problem management	<i>Service Problem Management processes are responsible for the management of problems associated with specific services. The objective of these processes is to respond immediately to reported service problems or failures in order to minimize their effects on customers, and to invoke the restoration of the service, or provide an alternate service as soon as possible.</i>

2.2. SID ABEs

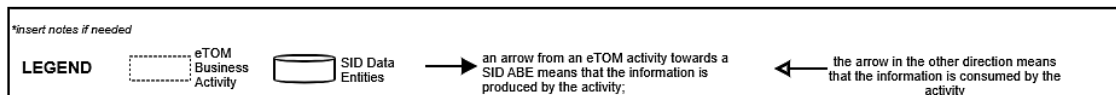
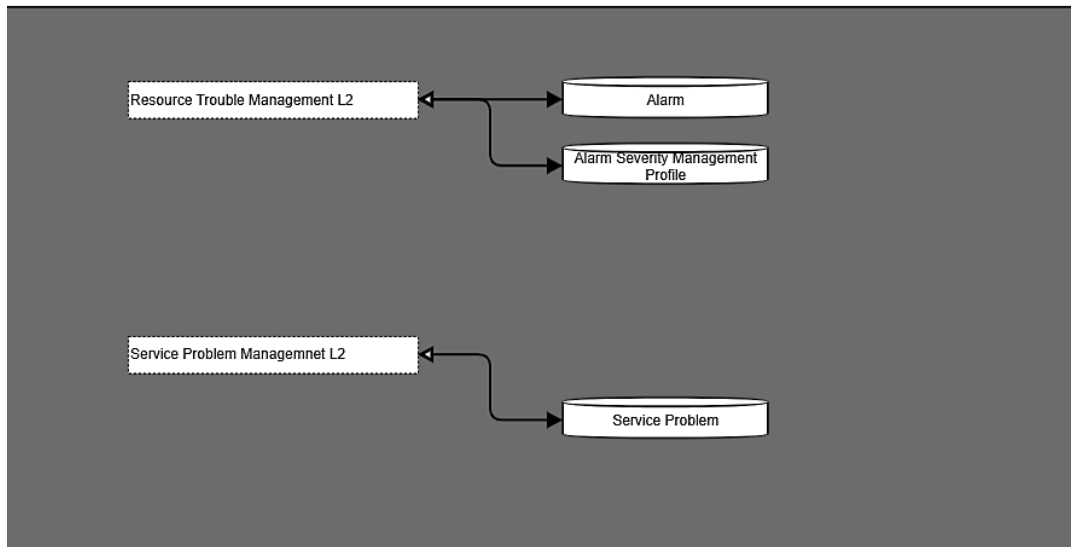
SID ABEs this ODA Component is responsible for:

SID ABE Level 1	SID ABE Level 2 (or set of BEs)*
Resource Trouble ABE	Resource Alarm ABE
Resource Trouble ABE	Alarm Severity Assignment Profile ABE
Service Problem ABE	

*: if SID ABE Level 2 is not specified this means that all the L2 business entities must be implemented, else the L2 SID ABE Level is specified.

2.3. eTOM L2 - SID ABEs links

eTOM L2 vS SID ABEs links for this ODA Component.



2.4. Functional Framework Functions

Function ID	Function Name	Function Description	Aggregate Function Level 1	Aggregate Function Level 2
5.6.2.4	Resource Trouble Analytics	Resource Trouble Cause Identification Function supports the trouble causes identification, that is to say the reasons why the installed resource is in trouble and among these reasons the root cause of the resource trouble.	Resource Trouble Management	Resource Trouble Analytics
5.6.2.2	Resource Trouble Resolution	Resource Trouble Resolution Function identifies the necessary resolution actions to solve the Resource Trouble according to Resource Problem Policy. It is based on diagnostic results.	Resource Trouble Management	Resource Trouble Resolution
4.6.1	Service Problem Management	Service Problem Management groups functions in charge of managing Service Trouble.	Service Problem Management	

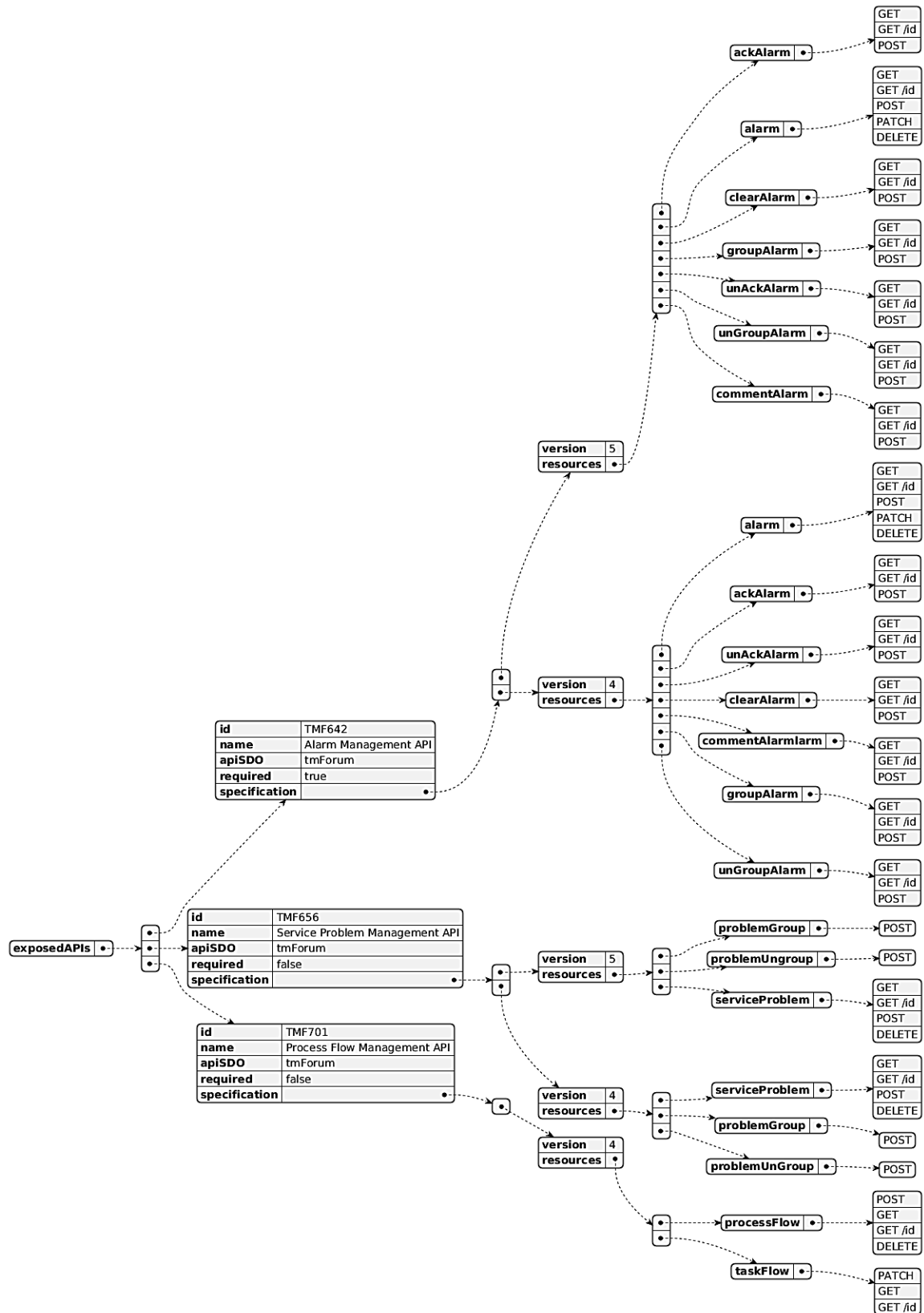
3. TM Forum Open APIs & Events

The following part covers the APIs and Events; This part is split in 3:

- List of **Exposed APIs** - This is the list of APIs available from this component.
- List of **Dependent APIs** - In order to satisfy the provided API, the component could require the usage of this set of required APIs.
- List of **Events (generated & consumed)** - The events which the component may generate is listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

3.1. Exposed APIs

Following diagram illustrates API/Resource/Operation:



API ID	API Name	Optionality	API Version	Operations	Rationale
TMF642	Alarm Management	mandatory	4	alarm: - POST - GET - GET /id - PATCH /id - DELETE /id ackAlarm: (optional) - o POST - o GET - o GET /id unAckAlarm: (optional) - o POST - o GET - o GET /id clearAlarm: - o POST - o GET - o GET /id commentAlarm: (optional) - o POST - o GET - o GET /id groupAlarm: (optional) - o POST - o GET - o GET /id unGroupAlarm: (optional) - o POST - o GET - o GET /id	
			5	alarm: - POST - GET - GET /id - PATCH /id - DELETE /id ackAlarm: (optional) - o POST - o GET - o GET /id unAckAlarm: (optional) - o POST - o GET - o GET /id clearAlarm: - o POST - o GET - o GET /id commentAlarm: (optional)	

API ID	API Name	Optionality	API Version	Operations	Rationale
				○ POST ○ GET ○ GET /id groupAlarm: (optional) ○ POST ○ GET ○ GET /id unGroupAlarm: (optional) ○ POST ○ GET ○ GET /id	
TMF656	Service Problem Management	optional	4	serviceProblem: - POST - GET - GET /id - PATCH /id - DELETE /id serviceProblemEventRecord: ○ GET ○ GET /id problemGroup: ○ Post problemUnGroup: ○ Post	
			5	serviceProblem: - POST - GET - GET /id - PATCH /id - DELETE /id serviceProblemEventRecord: ○ GET ○ GET /id problemGroup: ○ Post problemUnGroup: ○ Post	
TMF701	Process Flow	4	optional	processFlow: • GET • GET /id • POST • DELETE /id taskFlow: • GET • GET /id • PATCH /id	

3.2. Dependent APIs

Following diagram illustrates API/Resource/Operation potentially used by the Fault Management component:

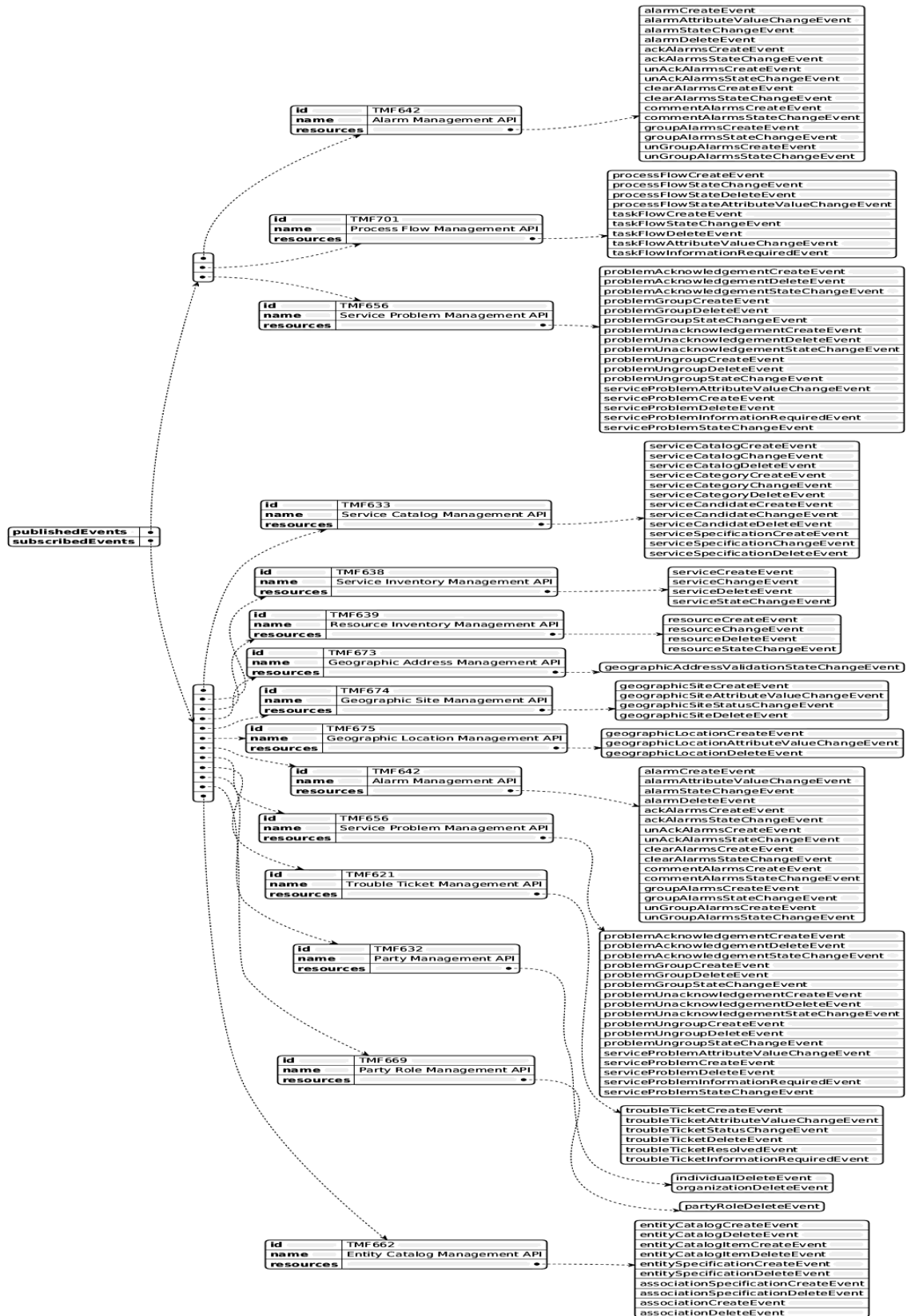
Your data does not sound like YAML data

API ID	API Name	API Version	Mandatory / Optional	Operations	Rationale
TMF633	Service Catalogue	4	optional	serviceCatalogue: - GET /id - GET serviceCategory: - GET /id - GET serviceCandidate: - GET /id - GET serviceSpecification: - GET /id - GET	
TMF634	Resource Catalogue	4.1	optional	resourceCatalogue: - GET /id - GET resourceCategory: - GET /id - GET resourceCandidate: - GET /id - GET resourceSpecification: - GET /id - GET	
TMF638	Service Inventory	4	optional	service: - GET /id - GET	
TMF639	Resource Inventory	4	optional	resource: - GET /id - GET	
TMF673	Geographic Address Management	4	optional	geographicAddress: - GET - GET /id geographicSubAddress: - GET - GET /id	
TMF674	Geographic Site	4	optional	geographicSite: - GET - GET /id	
TMF675	Geographic Location	4	optional	geographicLocation: - GET - GET /id	

API ID	API Name	API Version	Mandatory / Optional	Operations	Rationale
				<i>retrieveGeographicLocation:</i> - GET - GET /id	
TMF642	Alarm Management	4	optional	<i>alarm:</i> - GET - GET /id	Included to support alarm federation architectures.
TMF656	Service Problem Management	4	optional	<i>serviceProblem:</i> - GET - GET /id	Included to support alarm federation architectures.
TMF621	Trouble Ticket	4	optional	<i>troubleTicket:</i> - GET - GET /id	
TMF632	Party	4	optional	<i>individual:</i> - GET - GET /id <i>organization:</i> - GET - GET /id	The ownership can be achieved by using Party or by using alarm attributes in the alarm ABEmodel.
TMF669	Party Role	4	optional	<i>partyRole:</i> - GET - GET /id	
TMF662	Entity Catalogue	4	optional	<i>entityCatalogue:</i> - GET /id - GET <i>entityCategory:</i> - GET /id - GET <i>entityCatalogItem</i> - GET /id - GET <i>entitySpecification:</i> - GET /id - GET	

3.3. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections:



4. Machine Readable Component Specification

Refer to the ODA Component Directory for the machine-readable component specification file for this component.

5. References

5.1. TMF Standards related versions

Standard	Version(s)
SID	24.5
eTOM	24.5
Functional Framework	24.5

5.2. Jira References

5.2.1. SID

[\[ISA-1288\] Missing Network Trouble Ticket Entity - TM Forum JIRA](#)

5.3. Further resources

1. IG1228: please refer to IG1228 for defined use cases with ODA components interactions.

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
v1.0.0	11-Feb-2025	Yuval Stein	First Version
v1.0.0	19-May-2025	Rosie Wilson	Final Administrative Updates
v1.0.1	09-Jul-2025	Rosie Wilson	Updates to description per agreed YAML updated on 03 Jun 2025 Final Administrative Updates
v1.1.0	25-Jun-2026	Sannah Sibaya	Final edits prior to publication

6.1.2. Release History

Release Status	Date Modified	Modified by:	Description of changes
Pre-production	19-May-2025	Rosie Wilson	First Version
Pre-production	23-Jun-2025	Adrienne Walcott	Updated to Member Evaluated status
Pre-production	09-Jul-2025	Rosie Wilson	New release 1.0.1
Production			
Pre-production	25-Jun-2026	Sannah Sibaya	New Release 1.1.0

6.2. Acknowledgements

This document was prepared by the members of the TM Forum Component and Canvas project team:

Team Member (@mention)	Company	Role*
Yuval Stein	Amdocs	Author
Vance Shipley	SigScale	Contributor
Milind Bhagwat	BT	Contributor
Gaetano Biancardi	Accenture	Reviewer
Sylvie Demarest	Orange	Reviewer
Emmanuel A. Otchere	Huawei	Reviewer
Anastasios Sarantis	City Fibre	Reviewer
Abinash Vishwakarma	Netcracker	Reviewer
Jose Macaluso	Xacria	Reviewer
Kamal Maghsoudlou	Ericsson	Reviewer
Milind Bhagwat	BT	Reviewer
Ritu Arora	BT	Reviewer
Matteo Destino	Accenture	Reviewer

*Select from: Project Chair, Project Co-Chair, Author, Editor, Key Contributor, Additional Input, Reviewer